

Information and Communication Technology

Chapter 1: Foundations and Applications

Learning Objectives

- Identify data and information.
- Understand what a system is.
- Identify systems in day-to-day activities.
- Understand the characteristics of quality information.
- Explore various applications of ICT.
- Trace the evolution of the computer through its generations.

1.1 Data and Information

Numbers, words, images, and symbols which do not bear a meaning when standing alone are called **data**. While we can arrive at meaningful **information** by arranging and processing data, we can use them to make decisions as well.

Submitting data for processing is called **"Input"** and the result we get after processing is called **"Output."** The collection of all these components is a **"System."** Hence, the purpose of a system is to receive data, process and store them, and provide the results when required.

Example: Student Performance Data

Raw data: Ravi 78, 90, 79, 67, 76, 98; Rizwan 87, 70, 80, 75, 80, 80; Krishan 76, 78, 67, 80, 79, 76.

Name	Language	Maths	Science	History	Health	English	Total	Average	Rank
Rizwan	87	70	80	75	80	80	472	78.66	1
Ravi	78	90	79	67	76	78	468	78.00	2
Saman	76	78	67	80	79	76	456	76.00	3

1.2 The System

A system is a combination of components that work together to achieve a specific goal. In a computer system, this involves:

- **Input:** Providing raw data to the system.
- **Process:** Transforming data into meaningful information.
- **Output:** The result of the processing.
- **Storage:** Retaining data and information for future use.

1.3 Characteristics of Quality Information

Information is only valuable if it is of high quality. The five key characteristics are:

- **Relevancy:** Information must be suitable for the intended purpose.
- **Completeness:** It must contain all the necessary parts.
- **Accuracy:** It must be correct and reliable.
- **Timeliness:** It must be available when it is needed.
- **Low Cost:** The cost of obtaining information should be less than the benefit it provides.

1.4 Applications of ICT

e-Government

Using ICT to provide government services to citizens (G2C), businesses (G2B), and other government entities (G2G). Examples include utility bill payments, vehicle license renewals, and registration of businesses.

Education

ICT acts as a major teaching aid. Tools include Learning Management Systems (LMS), Computer Aided Learning (CAL), and Web-Based Training (WBT), allowing students to learn remotely and interactively.

Health Sector

Advanced medical equipment relies on ICT for diagnosis and treatment:

- **CAT (Computerized Axial Tomography):** Creates 3D images of body organs.
- **MRI (Magnetic Resonance Imaging):** Uses magnetic fields to image internal structures.

- **ECG (Electrocardiogram):** Monitors heart electrical activity.
- **Telemedicine & Telesurgery:** Remote consultation and surgery using high-speed connectivity.

Agriculture & Business

In agriculture, sensors monitor soil and weather, and RFID tracks livestock. In business, e-commerce, ATMs, and electronic Point of Sale (ePOS) systems have transformed transactions.

1.5 Evolution of the Computer

Generation	Period	Hardware Technology	Key Features
First	1940–1956	Vacuum Tubes	Massive size, slow, high heat, low reliability.
Second	1956–1963	Transistors	Smaller, faster, and more energy-efficient.
Third	1964–1971	Integrated Circuits (ICs)	Introduction of keyboards, monitors, and OS.
Fourth	1971–1989	LSI and VLSI	Microprocessors, portable computers (laptops).
Fifth	1989–Present	ULSI and AI	Artificial Intelligence, voice/handwriting recognition.

Summary

Data consists of raw symbols that lack context. Information is the result of processing data to make it useful for decision-making. ICT encompasses the technology used to manage and exchange this information. Computers have evolved through five distinct generations, shifting from large vacuum tubes to modern AI-driven processors.